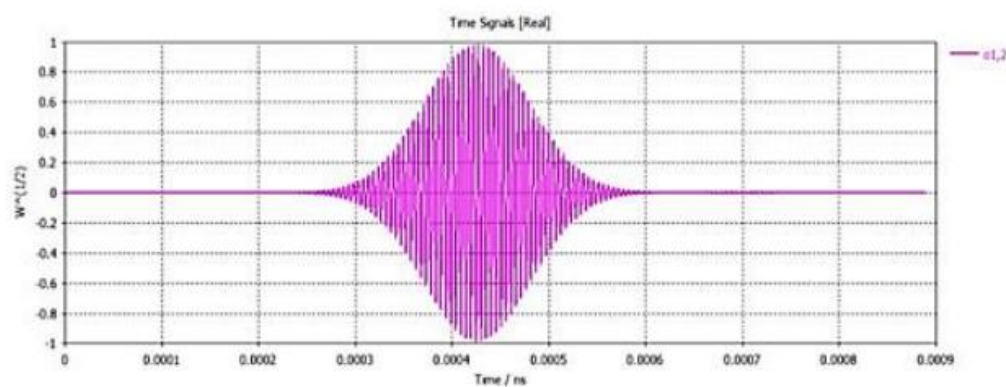
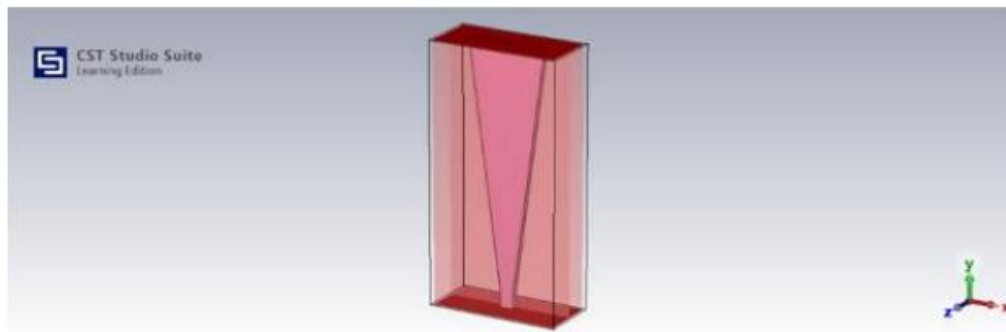


EXNO: 12 Performance Evaluation of a Linear Tapered Rectangular Waveguide in terms of power distribution analysis per ports and S-parameters.

Test Case 1: Simulate a linear tapered Rectangular Waveguide using CST Studio Suite, analyze its electromagnetic behavior, and examine the power distribution among ports over a specified frequency range.

Simulation Results:

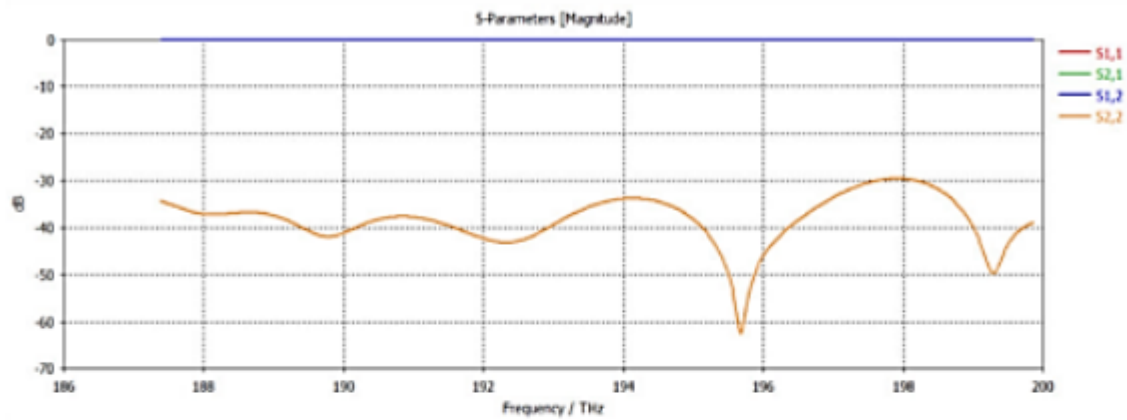


Output:

Port (i-input, o-output)	Time (ns)	Peak Power $W^{1/2}$
O1,2	0.00042	1.00
O2,1	0.00042	1.00
O1,1	0.00030	0.025
O2,2	0.00030	0.023

Test Case 2: Simulate a linear tapered Rectangular Waveguide using CST Studio Suite, analyze its electromagnetic behavior, and examine the S parameters.

Simulation Results:



Output:

Frequency (GHz)	S11 (dB)	S21 (dB)	S12 (dB)	S22 (dB)
187	0	0	0	-35
190	0	0	0	-41
192	0	0	0	-43
194	0	0	0	-34
196	0	0	0	-61
198	0	0	0	-30
200	0	0	0	-40